

Intelligent Circuit Identification Framework for Automated Simulation of Power Equipment with Winding Structure: From Circuit Diagram to FRA

Qiwen Ye, Yu Chen, Zong Deng, Renhe Shao, Zhongyong Zhao, *Member, IEEE*, Xiaoqing Ji, Yuxuan Ding, Zhengzheng Liu

Abstract—Frequency response analysis (FRA) is a well-established technique for evaluating the condition of power equipment with winding structures. Accurate circuit-based modeling helps researchers understand fault mechanisms through FRA signatures, offering a reliable basis for fault diagnosis. However, conventional modeling approaches often require manual conversion of circuit topologies into mathematical representations, a process that is labor-intensive, error-prone, and relies on specialized expertise. Therefore, this study proposes an intelligent circuit identification framework that automates the derivation of mathematical representations from circuit diagrams to FRA signatures. Specifically, the proposed framework includes a six-step process: component detection, unit recognition and component numbering, branch detection and node identification, incidence matrix construction, Kirchhoff equation generation, and state-space modeling. By integrating deep learning (DL), graph theory algorithms, and circuit theory principles, it systematically transforms circuit diagrams into state-space models. Furthermore, the generated state-space models can be combined with different optimization algorithms for parameter identification. The effectiveness of the proposed framework is validated through three case studies: a fifth-order RLC lumped parameter network modeled in Simulink, a practical synchronous machine stator winding with two different equivalent circuit topologies.

Index Terms—Automated modeling, circuit identification, winding modeling, frequency response analysis (FRA), frequency-domain simulation, state-space model.

I. INTRODUCTION

FREQUENCY response analysis (FRA) is a common technique for evaluating the condition of power equipment with winding structures, such as transformers and synchronous machines, by comparing measured and reference FRA signatures [1]–[4]. Accurate equivalent circuit models are funda-

mental for researchers and maintenance personnel to understand fault mechanisms through FRA signatures, providing a reliable foundation for fault diagnosis [5]–[8]. Furthermore, these models can support various downstream tasks, including parameter identification, transient analysis, and frequency-domain analysis [2], [3], [9]–[12].

Traditional equivalent circuit models for windings in power equipment are generally categorized into white-box, black-box, and gray-box models [13]–[16]. White-box methods, such as the finite element method (FEM), rely on physics-based laws to extract circuit parameters but often require solving complex partial differential equations with high computational cost [13], [14]. Black-box methods extract circuit parameters by fitting external measurements and enable rapid end-to-end numerical simulations, but they usually lack physical interpretability [15]. Gray-box methods combine the advantages of both approaches by using predefined circuit models with parameters identified from measurements or empirical data [2], [16].

In circuit-based FRA modeling, equivalent circuit topologies are commonly constructed by referring to models reported in the literature or by adapting them according to measured resonance characteristics [2], [17], [18]. A large number of such equivalent circuit diagrams are available for transformers, synchronous machines, and other winding-based power equipment [2], [9], [17]–[19]. Even in white-box modeling, circuit representations are often required after equivalent parameters are extracted from FEM or electromagnetic-field calculations, so that the frequency response can be calculated [13], [14]. However, once a circuit topology is selected, it must still be converted into mathematical representations before frequency-response calculation and parameter identification can be performed. In existing studies [2], [16], [19], [20], this conversion is typically performed manually using Kirchhoff's Current Law (KCL), Kirchhoff's Voltage Law (KVL), and state-space formulation. This process is tedious, time-consuming, and prone to human error, especially when the circuit order increases, or different topologies need to be compared [21], [22].

In recent years, artificial intelligence (AI) has been increasingly applied to the power system domain [23]–[26], offering feasible methods to address these challenges. Notably, deep learning (DL) techniques in image recognition have demonstrated potential for automatic circuit diagram recognition. Once circuit topology information is extracted

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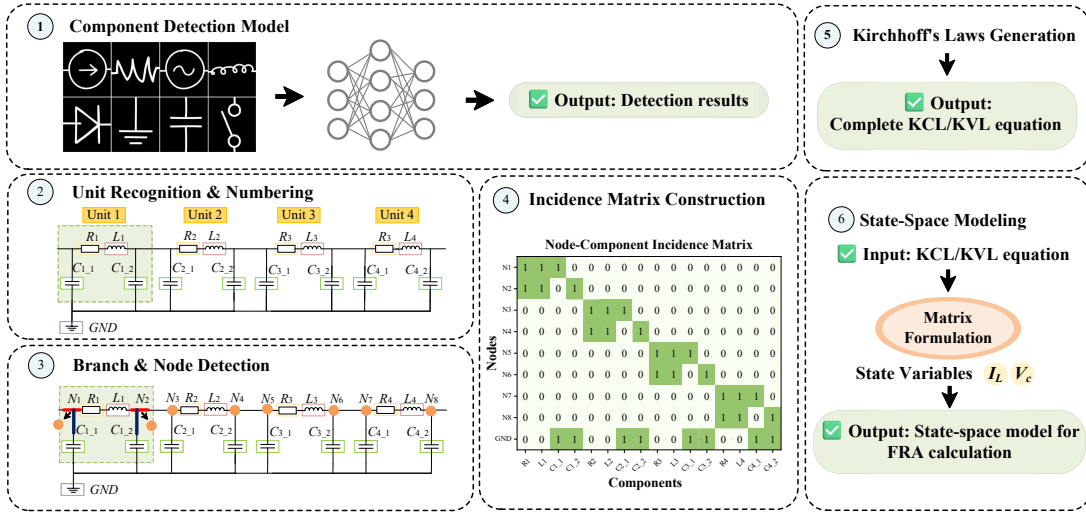


Fig. 1. Overview of the six-step intelligent circuit identification framework.

from a circuit diagram, it can be further utilized to automate the generation of mathematical representations. The generated mathematical representations can be extended to diverse FRA-based applications [1], [5], [9], [27].

Although AI-based circuit analysis has attracted increasing attention, existing studies remain fragmented. For instance, Ref. [28] investigates the digitization of schematic diagrams, enabling conversion from images to SPICE netlists, though primarily for basic simulation verification. Ref. [29] introduces the AEM-PCB Reverser (Aesthetic Evaluation Metric Printed Circuit Board Reverser) system, which employs reinforcement learning to generate schematic diagrams from PCBs, demonstrating the potential of AI for circuit recognition but mainly serving PCB design verification. In component recognition, Ref. [30] proposes an ORB (Oriented FAST and Rotated BRIEF)-based method for identifying hand-drawn electronic components, focusing on digitization as a preliminary step for schematic processing. Ref. [31] presents the PowerVision framework for schematic recognition and netlist generation, validated through LTspice simulations and primarily targeting power-electronics design and analysis. Recently, Ref. [32] proposed a method that combines component detection, connectivity inference, and optical character recognition (OCR) to convert circuit schematic images into SPICE netlists, mainly focusing on image-to-netlist conversion. These studies demonstrate the feasibility of using AI techniques for circuit diagram processing. However, existing studies mainly address separate tasks, such as component recognition, schematic digitization, and netlist generation. These tasks are more related to computer vision and image processing than to practical electrical engineering applications. A practical framework that further uses the recognized circuit information for specific engineering tasks is still lacking.

Therefore, this study proposes an intelligent circuit identification framework that automates the derivation of mathematical representations from circuit diagrams to FRA signatures. The main contributions are as follows:

- 1) An intelligent circuit identification framework is pro-

posed for two-port RLC networks of winding-based power equipment. Unlike general AI circuit recognition tools limited to component detection and netlist generation, it incorporates physics priors to extract complete topology from diagrams, enabling end-to-end support for FRA modeling.

- 2) A graph-theory-based methodology is integrated to generate state-space models, automatically converting circuit topologies into mathematical formulations suitable for calculating FRA data.
- 3) The framework is combined with various optimization algorithms to achieve parameter identification, linking external measurements to internal circuit parameters.

This study is organized as follows: Section II introduces the proposed framework. Section III validates the framework through three case studies, followed by a discussion of limitations and future directions in Section IV. Section V concludes this study.

II. METHODOLOGY

This section introduces the proposed intelligent circuit identification framework, which detects components, extracts the circuit topology from circuit diagrams, and converts it into mathematical representations. The framework consists of six steps, as shown in Fig. 1.

A. Component Detection and Unit Recognition

Based on FRA, any windings can be represented as equivalent circuits composed of the resistor (R), inductor (L), and capacitor (C) elements, with the ladder network being the most widely adopted one [2], [9], [17], [18]. For automated modeling, schematic symbols must be accurately extracted from circuit diagrams. As illustrated in Fig. 2, component identification is performed using a detection model. The model employs the YOLOv11 architecture (Fig. 3), which takes circuit diagrams as input and outputs the detected component categories along with their bounding box coordinates. Training

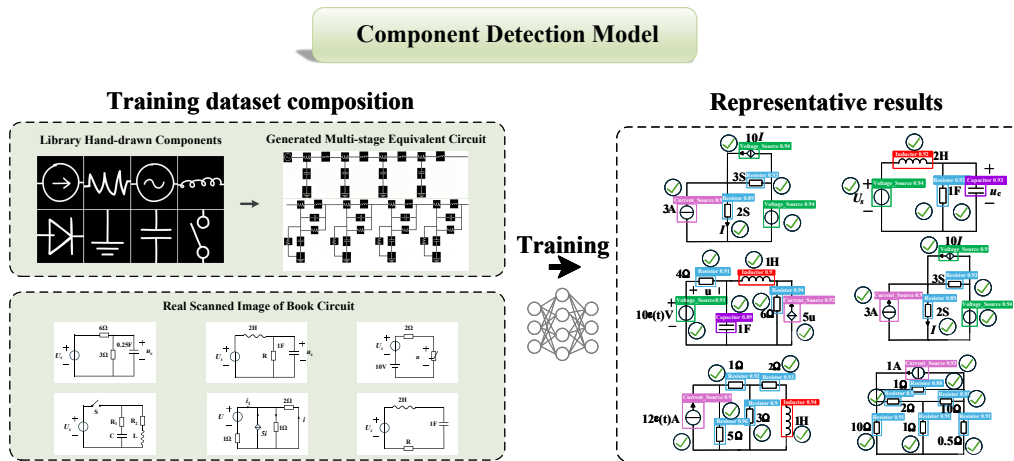


TABLE II
COMPONENT DETECTION PERFORMANCE ON TEST DATASET

Component	Images	Instances	Precision	Recall	mAP@0.5	mAP@0.5:0.95
Resistor	7	22	99.9%	100%	99.5%	74.8%
Capacitor	5	6	96.7%	100%	99.5%	60.2%
Inductor	5	6	93.2%	100%	99.5%	72.0%
Voltage Source	6	8	96.3%	100%	99.5%	99.5%
Current Source	2	3	96.2%	100%	99.5%	86.5%
Overall	15	45	96.5%	100%	99.5%	68.6%

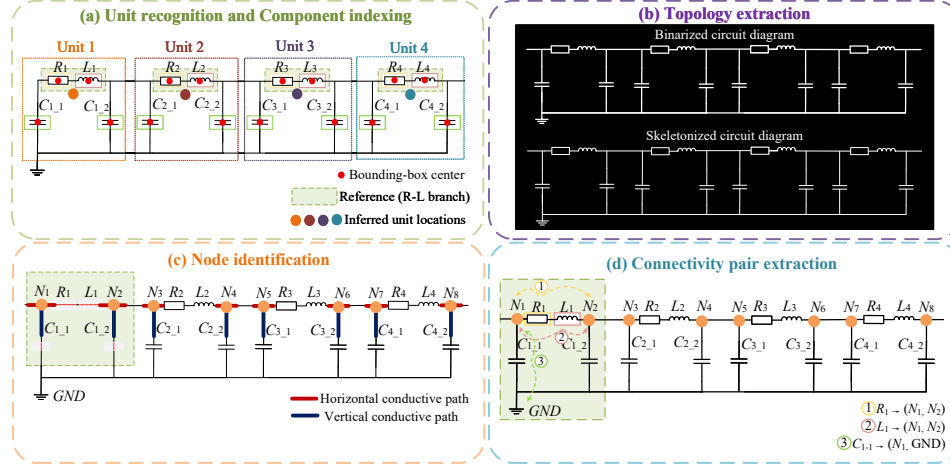


Fig. 4. Illustration of unit recognition, topology extraction, node identification, and connectivity pair extraction in the cascaded circuit. (a) unit recognition and component numbering. (b) topology extraction. (c) node identification. (d) connectivity pair extraction.

B. Topology Extraction and Incidence Matrix Construction

Following component detection and unit recognition, Step 3 and Step 4 extract the circuit topology and construct the incidence matrix to determine the connectivity relationships among the components. As illustrated in Fig. 4(b), the circuit diagrams are first binarized via Otsu adaptive thresholding [34] and subsequently reduced to single-pixel-width skeletons using Zhang-Suen skeletonization [35]. To isolate the conductive paths, the components detected in Section II-A are removed from the skeletonized diagram, with any broken segments reconnected to ensure conductor continuity. Finally, as shown in Fig. 4(c), circuit nodes are identified at the junctions where horizontal and vertical conductive paths intersect [28].

To enable automated model generation, the extracted topology information should be represented in a machine-readable form. For each detected component, the framework determines its two terminal nodes by analyzing the nearest connected nodes around its bounding box. The corresponding node pair is defined as the connectivity pair of this component, as shown in Fig. 4(d). The ground node is also treated as a common reference node, with multiple ground symbols regarded as the same GND node. The identified connectivity pairs are subsequently organized into a node-component incidence matrix, as shown in Fig. 5. The node-component incidence matrix provides the connectivity information required to generate circuit equations, while the component types and parameters determine the entries of the corresponding circuit matrices. Based on the extracted topology, Kirchhoff's laws and component constitutive relations are automatically formulated, as shown in Fig. 6.

Specifically, series-connected components between the same pair of nodes are assigned to a common branch current. For example, R_1 and L_1 share the branch current I_1 between N_1 and N_2 , which is used to generate the corresponding KVL equation. Nodes connected only by ideal conductors are merged as equipotential nodes. For instance, the conductor-connected nodes associated with N_2 are represented by the node voltage V_2 . The resulting independent node voltages are then used to generate the KCL equations.

Node-Component Incidence Matrix

Nodes	N_1	N_2	N_3	N_4	N_5	N_6	N_7	N_8	GND
Components	1	1	1	0	0	0	0	0	0
R_1	1	1	0	0	0	0	0	0	0
L_1	1	1	0	0	0	0	0	0	0
$C_{1,1}$	1	1	0	0	0	0	0	0	0
$C_{1,2}$	0	0	0	0	0	0	0	0	0
R_2	0	0	0	1	1	1	0	0	0
L_2	0	0	0	1	1	1	0	0	0
$C_{2,1}$	0	0	0	1	1	1	0	0	0
$C_{2,2}$	0	0	0	0	0	0	0	0	0
R_3	0	0	0	0	0	0	1	1	0
L_3	0	0	0	0	0	0	1	1	0
$C_{3,1}$	0	0	0	0	0	0	1	1	0
$C_{3,2}$	0	0	0	0	0	0	0	0	0
R_4	0	0	0	0	0	0	0	1	1
L_4	0	0	0	0	0	0	0	1	1
$C_{4,1}$	0	0	0	0	0	0	0	1	1
$C_{4,2}$	0	0	0	0	0	0	0	0	0

Fig. 5. Illustration of the node-component incidence matrix. Rows represent circuit nodes ($N_1, N_2, \dots, N_n, GND$), columns represent components, and matrix entries indicate connectivity (0 for no connection and 1 for connection).

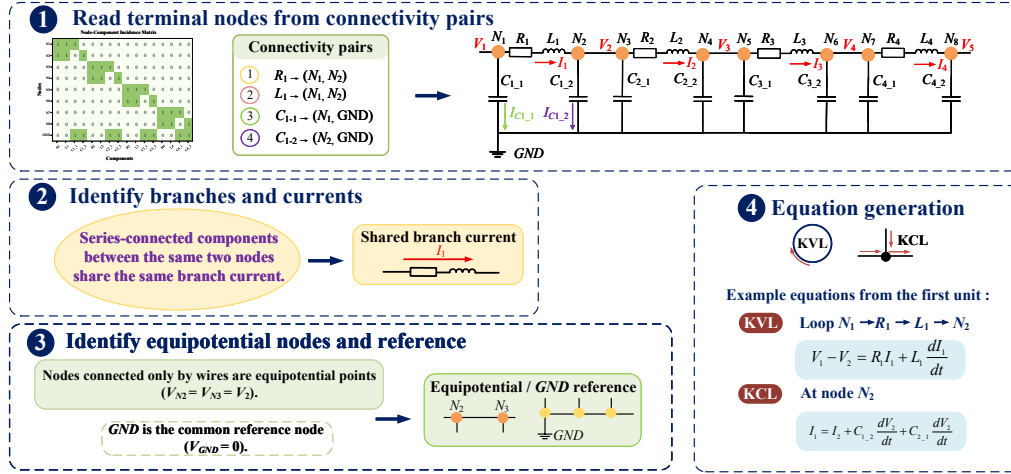


Fig. 6. Illustration of equation generation in the cascaded circuit.

C. Frequency-Domain State-Space Modeling

After component detection and topology extraction, the proposed framework converts the identified circuit topology into a mathematical representation (i.e., state-space model) for FRA calculation. To reduce redundant variables, series-connected components without intermediate branching nodes are assigned the same branch current, while nodes connected only by ideal conductors are merged as a single equipotential node. Thus, components or branches connected between the same pair of nodes share the same voltage difference. The generated KCL and KVL equations can then be organized as

$$\mathbf{H}\mathbf{I} = \mathbf{C}\dot{\mathbf{V}} + \mathbf{G}\mathbf{V} \quad (1)$$

$$\mathbf{T}\mathbf{V} = \mathbf{R}\mathbf{I} + \mathbf{L}\dot{\mathbf{I}} \quad (2)$$

where $\mathbf{H}$ and $\mathbf{T}$ are the current and voltage coefficient matrices, respectively. $\mathbf{R}$, $\mathbf{L}$, $\mathbf{C}$, and $\mathbf{G}$ denote the resistance, inductance, capacitance, and conductance matrices, respectively.

To calculate the terminal frequency response, the input voltage V_{in} is separated from the remaining node-voltage variables, defined as $\mathbf{V}_s = [V_2, \dots, V_n]^T$. Accordingly, (1) and (2) become:

$$\mathbf{H}\mathbf{I} = \mathbf{C}_0 \dot{V}_{in} + \mathbf{C}_s \dot{\mathbf{V}}_s + \mathbf{G}_0 V_{in} + \mathbf{G}_s \mathbf{V}_s \quad (3)$$

$$\mathbf{T}_0 V_{in} + \mathbf{T}_s \mathbf{V}_s = \mathbf{R}\mathbf{I} + \mathbf{L}\dot{\mathbf{I}} \quad (4)$$

where $\mathbf{T}_0$, $\mathbf{G}_0$, $\mathbf{C}_0$ are the first column vectors of $\mathbf{T}$, $\mathbf{G}$, and $\mathbf{C}$, respectively, and $\mathbf{T}_s$, $\mathbf{G}_s$, $\mathbf{C}_s$ are the remaining sub-matrices.

By transforming the formulas, the following matrix form is obtained:

$$\begin{pmatrix} \mathbf{C}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{L} \end{pmatrix} \begin{pmatrix} \dot{\mathbf{V}}_s \\ \dot{\mathbf{I}} \end{pmatrix} = \begin{pmatrix} -\mathbf{G}_s & \mathbf{H} \\ \mathbf{T}_s & -\mathbf{R} \end{pmatrix} \begin{pmatrix} \mathbf{V}_s \\ \mathbf{I} \end{pmatrix} + \begin{pmatrix} -\mathbf{G}_0 \\ \mathbf{T}_0 \end{pmatrix} V_{in} + \begin{pmatrix} -\mathbf{C}_0 \\ \mathbf{0} \end{pmatrix} \dot{V}_{in} \quad (5)$$

In the formula, the vectors and structural matrices are defined as $\mathbf{X} = \begin{pmatrix} \mathbf{V}_s \\ \mathbf{I} \end{pmatrix}$, $\mathbf{A} = \begin{pmatrix} \mathbf{C}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{L} \end{pmatrix}$, $\mathbf{P} = \begin{pmatrix} -\mathbf{G}_s & \mathbf{H} \\ \mathbf{T}_s & -\mathbf{R} \end{pmatrix}$, $\mathbf{Q} = \begin{pmatrix} -\mathbf{C}_0 \\ \mathbf{0} \end{pmatrix}$, and $\mathbf{B} = \begin{pmatrix} -\mathbf{G}_0 \\ \mathbf{T}_0 \end{pmatrix}$. The standard state-space model is then expressed as:

$$\mathbf{A}\dot{\mathbf{X}} = \mathbf{B}\mathbf{X} + \mathbf{P}V_{in} + \mathbf{Q}\dot{V}_{in} \quad (6)$$

The output variable is selected according to the terminal characteristic of interest:

$$\mathbf{Y} = \mathbf{M}\mathbf{X} \quad (7)$$

where $\mathbf{M}$ is the output-selection matrix. Applying the Laplace transform to (6) gives:

$$\mathcal{H}(s) = \frac{Y(s)}{V_{in}(s)} = \mathbf{M}(s\mathbf{A} - \mathbf{B})^{-1}(\mathbf{P} + s\mathbf{Q}) \quad (8)$$

By substituting $s = j2\pi f$, the frequency-domain response can be evaluated over the required frequency range. Depending on the selected input and output variables, the framework can obtain terminal impedance, terminal admittance, or voltage-gain response, expressed as:

$$Z(s) = \frac{V_{in}(s)}{I_{in}(s)} \quad Y_{in}(s) = \frac{I_{in}(s)}{V_{in}(s)} \quad G(s) = \frac{V_{out}(s)}{V_{in}(s)} \quad (9)$$

D. Parameter Identification

The generated state-space representation is further used for parameter identification. Let θ denote the vector of circuit parameters to be identified. In this study, the genetic algorithm (GA) [36], particle swarm optimization (PSO) [37], and grey wolf optimizer (GWO) [38] are employed to search for the optimal parameter vector.

The objective is to minimize the discrepancy between the reference and identified terminal characteristics over the selected frequency range. The terminal characteristic is denoted by $\mathcal{F}(s)$, which can represent terminal impedance, terminal admittance, or voltage-gain response:

$$\mathcal{F}(s) \in \{Z(s), Y_{in}(s), G(s)\}. \quad (10)$$

For a given parameter vector θ , the identified terminal characteristic $\mathcal{F}_{id}(f_i; \theta)$ is calculated from the generated state-space model. The fitness function is defined as:

$$J(\theta) = \sum_{i=1}^N \left| \frac{\mathcal{F}_{ref}(f_i) - \mathcal{F}_{id}(f_i; \theta)}{\mathcal{F}_{ref}(f_i)} \right|^2, \quad (11)$$

Algorithm 1 Intelligent Circuit Identification and Automated Modeling

Require: Circuit diagram, reference terminal characteristic $\mathcal{F}_{\text{ref}}$.
Ensure: State-space representation, identified parameter vector θ .

- 1: // **Phase 1: Topology Extraction**
- 2: Detect circuit components.
- 3: Construct the node-component incidence matrix.
- 4: // **Phase 2: Frequency-Domain State-Space Modeling**
- 5: Generate Kirchhoff's equations.
- 6: Transform the generated equations into matrix form:
 $\mathbf{H}\mathbf{I} = \mathbf{C}\mathbf{V} + \mathbf{G}\mathbf{V}, \mathbf{T}\mathbf{V} = \mathbf{R}\mathbf{I} + \mathbf{L}\mathbf{I}$.
- 7: Separate V_{in} from the state variables and obtain the descriptor state-space representation:
 $\mathbf{A}\mathbf{X} = \mathbf{B}\mathbf{X} + \mathbf{P}\mathbf{V}_{\text{in}} + \mathbf{Q}\dot{\mathbf{V}}_{\text{in}}$.
- 8: // **Phase 3: Parameter Identification**
- 9: **while** termination criteria not met **do**
- 10: Update the parameter vector θ using the optimization algorithm.
- 11: Compute the identified terminal characteristic $\mathcal{F}_{\text{id}}(f_i; \theta)$ from the generated state-space model.
- 12: Evaluate the fitness value:
 $J(\theta) = \sum_{i=1}^N \left| \frac{\mathcal{F}_{\text{ref}}(f_i) - \mathcal{F}_{\text{id}}(f_i; \theta)}{\mathcal{F}_{\text{ref}}(f_i)} \right|^2$.
- 13: **end while**
- 14: **return** Identified parameter vector θ .

where $\mathcal{F}_{\text{ref}}(f_i)$ and $\mathcal{F}_{\text{id}}(f_i; \theta)$ denote the reference and identified terminal characteristics at frequency f_i , respectively, and N is the number of frequency points. By selecting different forms of $\mathcal{F}(s)$, the same identification framework can be applied to impedance, admittance, or voltage-gain fitting tasks. The overall procedure of the proposed framework is summarized in Algorithm 1.

III. EXPERIMENTAL VALIDATION

To validate the proposed intelligent circuit identification framework for parameter identification, three case studies are presented in this section.

A. Simulation Validation: Fifth-Order RLC Lumped Parameter Network in Simulink

A fifth-order RLC lumped parameter network is used as a representative equivalent circuit to validate the proposed framework. Such networks are commonly adopted to approximate the distributed characteristics of power equipment with winding structures, such as synchronous machines and transformers, especially in broadband frequency modeling [39], [40]. As shown in Fig. 7, the circuit consists of five cascaded RLC units. Each unit contains a series resistance R_i , a series inductance L_i , and a ground capacitance C_i , while adjacent units are coupled through capacitances K_i .

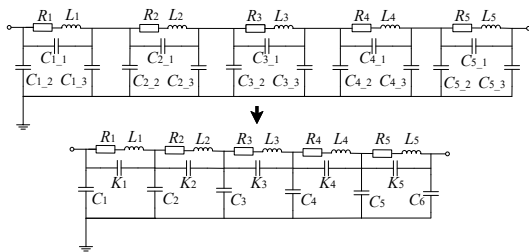


Fig. 7. Fifth-order RLC lumped parameter network.

The circuit diagram is first processed by the proposed identification framework. Component detection and unit recognition

are performed to identify the circuit elements and assign component labels. Branch detection and node identification are then conducted to extract the connectivity relationships, as shown in Fig. 8. Based on the identified nodes and components, the node-component incidence matrix is constructed, as shown in Fig. 9. This matrix provides the topological information required for automated equation generation and state-space modeling.

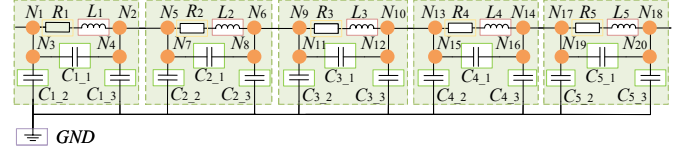


Fig. 8. Branch detection and node numbering for the fifth-order RLC network.

Node-Component Incidence Matrix

Nodes	N1	N2	N3	N4	N5	N6	N7	N8	N9	N10	N11	N12	N13	N14	N15	N16	N17	N18	N19	N20	GND
	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0
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Components

Fig. 9. Incidence matrix of the fifth-order RLC network.

Before equation generation, the detected component labels are mapped to the parameter labels used in the equivalent circuit. For the i -th unit, C_{i-1} is denoted as the coupling capacitance K_i . The first ground capacitance C_{1-2} is denoted as C_1 , while adjacent ground capacitances connected to the same equipotential node are merged into an equivalent capacitance. For example, C_{1-3} and C_{2-2} are represented by C_2 . Based on the extracted topology and the mapped parameter labels, the Kirchhoff equations for the network are generated as:

$$V_i - V_{i+1} = R_i I_i + L_i \dot{I}_i \quad (12)$$

$$I_i - I_{i+1} = K_i (\dot{V}_{i+1} - \dot{V}_i) + C_{i+1} \dot{V}_{i+1} + K_{i+1} (\dot{V}_{i+1} - \dot{V}_{i+2}) \quad (13)$$

where V_i denotes the voltage across C_i , and I_i denotes the current through L_i . The parameters R_i , L_i , and C_i represent the resistance, inductance, and equivalent ground capacitance of the i -th unit, respectively, while K_i represents the coupling capacitance between adjacent units.

Following the procedure in Section II-C, the generated Kirchhoff equations are transformed into the state-space

model. The corresponding parametric matrices are obtained as:

$$\mathbf{R} = \begin{bmatrix} 0 & R_1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & R_2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & R_3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & R_4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & R_5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{L} = \begin{bmatrix} 0 & L_1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & L_2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & L_3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & L_4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & L_5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{T} = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} K_1+C_1 & -K_1 & 0 & 0 & 0 & 0 & 0 \\ -K_1 & K_1+K_2+C_2 & -K_2 & 0 & 0 & 0 & 0 \\ 0 & -K_2 & K_2+K_3+C_3 & -K_3 & 0 & 0 & 0 \\ 0 & 0 & -K_3 & K_3+K_4+C_4 & -K_4 & 0 & 0 \\ 0 & 0 & 0 & -K_4 & K_4+K_5+C_5 & -K_5 & 0 \\ 0 & 0 & 0 & 0 & -K_5 & K_5+C_6 & 0 \end{bmatrix} \quad (14)$$

The generated parametric matrices are substituted into the descriptor state-space model to calculate the terminal response. In this case, the terminal impedance is selected as the target characteristic, i.e., $\mathcal{F}(s) = Z(s)$, followed by constructing a fitness function (11) to minimize the discrepancy between the reference and identified impedance responses. The reference circuit is constructed in Simulink with the preset parameters $R_i = 1.2 \Omega$, $L_i = 180 \mu\text{H}$, $C_i = 3 \text{ nF}$, and $K_i = 0.013 \text{ nF}$. The reference impedance $\mathcal{F}_{\text{ref}}(f)$ is extracted over the frequency range from 1 kHz to 1 MHz. Although the five circuit units are assigned identical preset values, their parameters are treated as independent variables during identification.

The identified parameters obtained by GA, PSO, and GWO are summarized in Table III. To evaluate the fitting performance, three metrics are used [2]: the mean absolute percentage error E_{MAPE} , the root mean square error E_{RMSE} , and the resonance-point error E_{RP} . The first two metrics evaluate the overall agreement over the full frequency range, while E_{RP} evaluates the discrepancy at selected resonance and anti-resonance points. They are defined as:

$$E_{\text{MAPE}}(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^N \frac{|\mathcal{F}_{\text{ref}}(f_i)| - |\mathcal{F}_{\text{id}}(f_i; \boldsymbol{\theta})|}{|\mathcal{F}_{\text{ref}}(f_i)|} \times 100\% \quad (15)$$

$$E_{\text{RMSE}}(\boldsymbol{\theta}) = \sqrt{\frac{1}{N} \sum_{i=1}^N (|\mathcal{F}_{\text{ref}}(f_i)| - |\mathcal{F}_{\text{id}}(f_i; \boldsymbol{\theta})|)^2} \quad (16)$$

$$E_{\text{RP}}(\boldsymbol{\theta}) = \sum_{i=1}^{N_{\text{RP}}} \left(\frac{|\mathcal{F}_{\text{ref}}(f_i^{\text{RP}})| - |\mathcal{F}_{\text{id}}(f_i^{\text{RP}}; \boldsymbol{\theta})|}{|\mathcal{F}_{\text{ref}}(f_i^{\text{RP}})|} \right)^2 \quad (17)$$

where $\mathcal{F}_{\text{ref}}(f_i)$ is the reference terminal characteristic, and $\mathcal{F}_{\text{id}}(f_i; \boldsymbol{\theta})$ is the identified terminal characteristic calculated using the parameter vector $\boldsymbol{\theta}$. N is the number of frequency points, f_i^{RP} denotes the selected resonance or anti-resonance frequency, and N_{RP} is the number of selected resonance and anti-resonance points.

The quantitative evaluation results are listed in Table IV, and the corresponding FRA curves are shown in Fig. 10. Among the three algorithms, GWO achieves the lowest E_{MAPE} and E_{RMSE} , with values of 2.6998% and 1.2486×10^{-1} , respectively. GA gives the lowest E_{RP} , with a value of 2.4277×10^{-3} at the selected resonance and anti-resonance points.

TABLE III
IDENTIFICATION RESULTS OF THE FIFTH-ORDER RLC NETWORK

Parameters	Preset Value	GA	PSO	GWO
$R_1 (\Omega)$	1.2000	1.2245	1.2437	1.2437
$R_2 (\Omega)$	1.2000	1.1892	1.2079	1.2330
$R_3 (\Omega)$	1.2000	1.2056	1.2145	1.2390
$R_4 (\Omega)$	1.2000	1.2133	1.2449	1.2403
$R_5 (\Omega)$	1.2000	1.1921	1.2013	1.2452
$L_1 (\text{mH})$	0.1800	0.1834	0.1873	0.1773
$L_2 (\text{mH})$	0.1800	0.1765	0.1825	0.1812
$L_3 (\text{mH})$	0.1800	0.1812	0.1733	0.1865
$L_4 (\text{mH})$	0.1800	0.1789	0.1766	0.1830
$L_5 (\text{mH})$	0.1800	0.1821	0.1825	0.1736
$K_1 (\text{nF})$	0.0130	0.0129	0.0126	0.0130
$K_2 (\text{nF})$	0.0130	0.0131	0.0136	0.0124
$K_3 (\text{nF})$	0.0130	0.0130	0.0134	0.0132
$K_4 (\text{nF})$	0.0130	0.0133	0.0135	0.0135
$K_5 (\text{nF})$	0.0130	0.0131	0.0134	0.0133
$C_1 (\text{nF})$	3.0000	3.1200	2.9300	3.0616
$C_2 (\text{nF})$	6.0000	5.8500	5.7167	6.2755
$C_3 (\text{nF})$	6.0000	6.1200	6.2804	6.1038
$C_4 (\text{nF})$	6.0000	5.9400	6.0707	6.2246
$C_5 (\text{nF})$	6.0000	6.0800	6.2946	5.8325
$C_6 (\text{nF})$	3.0000	2.8900	3.0274	2.9345

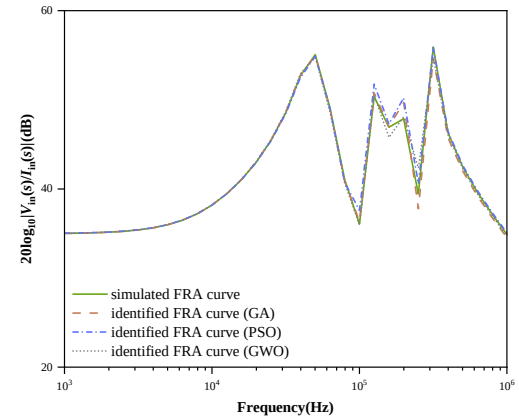


Fig. 10. Comparison between identified and simulated FRA signatures for the fifth-order RLC network.

B. Experimental Validation: Ladder Network Model of the Synchronous Machine Stator Winding

To further validate the proposed framework, experiments are performed on the stator winding of a 5 kW, 380 V, 9.5 A, 1500 r/min three-phase salient-pole synchronous machine. The experimental setup and measurement procedure follow [18], as shown in Fig. 11. An SFRB-R FRA analyzer, with a frequency-sweep accuracy of less than 0.01% and an amplitude accuracy of less than $\pm 1 \text{ dB}$, is used to measure the FRA signatures. The U-phase winding is selected as the representative case. A 10 V sinusoidal sweep signal from 1 kHz to 1 MHz is injected at the neutral point, and the response is measured at the U-phase terminal. In practical winding structures, magnetic coupling between different circuit units is

TABLE IV
QUANTITATIVE EVALUATION OF IDENTIFICATION PERFORMANCE FOR THE FIFTH-ORDER RLC NETWORK

Algorithm	$E_{\text{MAPE}} (\%)$	E_{RMSE}	E_{RP}
GA	3.4540	1.3521×10^{-1}	2.4277×10^{-3}
PSO	2.9717	1.2754×10^{-1}	3.4307×10^{-3}
GWO	2.6998	1.2486×10^{-1}	5.9507×10^{-3}

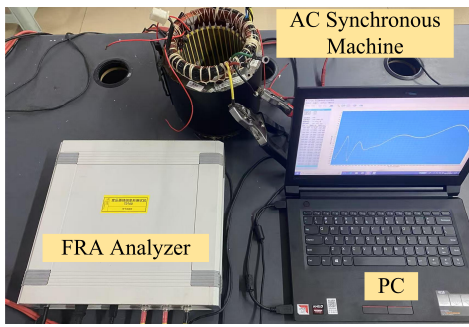


Fig. 11. Experimental setup for FRA measurement.

incorporated after all inductors are identified. Since windings typically exhibit coaxial configurations, their mutual inductance cannot be neglected [18]. Once all inductance components in the circuit diagram are determined, their mutual coupling relationships are also established. Specifically, mutual-inductance terms are placed in the off-diagonal entries of the inductance matrix, and the voltage equation of the i -th inductor includes the induced-voltage term:

$$\sum_{k \neq i} M_{ik} \frac{dI_k}{dt}. \quad (18)$$

Mutual inductance is constrained by physical coupling relationships, which should be considered in parameter identification, thereby reducing the search space and speeding up this process.

$$M_{ik} < L_{ii}, \quad M_{ik} < L_{kk}, \quad (19)$$

$$M_{ij} > M_{mn}, \quad \text{if } |i - j| < |m - n|. \quad (20)$$

The measured FRA signature of the U-phase winding is shown in Fig. 12. Two dominant resonance points are observed within the frequency range from 1 kHz to 1 MHz. In the low-frequency region, the inductive effect dominates, resulting in a decreasing response. In the high-frequency region, the capacitive effect becomes more pronounced, leading to oscillatory behavior. According to Refs. [17], [18], the order of the equivalent ladder network is determined as:

$$N_s = 2N - 1 \quad (21)$$

where N_s is the number of equivalent circuit sections and N is the number of dominant resonances. Since two dominant resonances are observed in this case, a third-order ladder network is adopted, as shown in Fig. 13.

The third-order ladder network is used as the input to the proposed framework. Fig. 14 shows the component detection and node numbering results, and Fig. 15 gives the corresponding node-component incidence matrix. Before equation generation, the detected component labels are mapped to the parameter labels in Fig. 13. Specifically, for $i = 1, 2, 3$, C_{i-1} is denoted as the same coupling capacitance K_i , and R_i is denoted as the stator resistance R_{si} . Adjacent capacitances connected to equipotential points are merged into an equivalent capacitance. For example, C_{1-3} and C_{2-2} are combined and represented by C_2 .

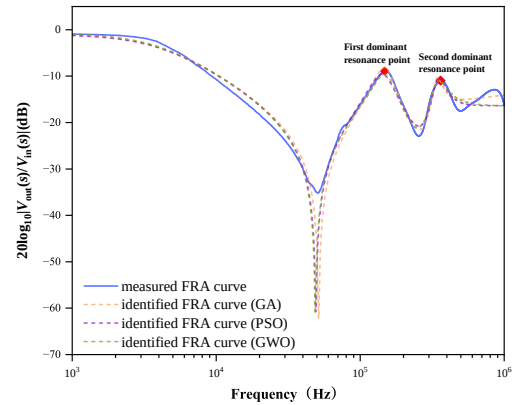


Fig. 12. Comparison between identified and measured FRA signatures for synchronous machine stator winding.

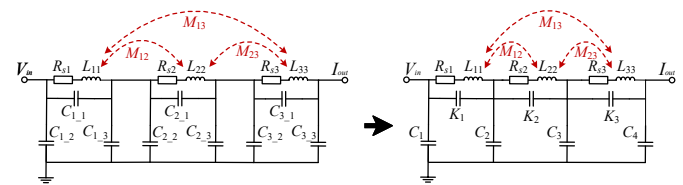


Fig. 13. A third-order ladder network, where M_{ik} denotes the mutual inductance between the i -th and k -th inductors.

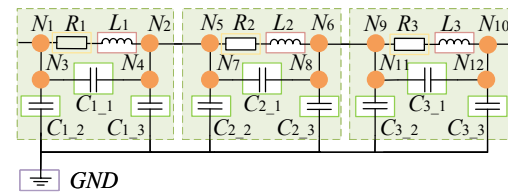


Fig. 14. Component detection and node numbering for the third-order ladder network.

Nodes	R1	L1	C1-1	C1-2	C1-3	R2	L2	C2-1	C2-2	C2-3	R3	L3	C3-1	C3-2	C3-3
N1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0
N2	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0
N3	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0
N4	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0
N5	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0
N6	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0
N7	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0
N8	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0
N9	0	0	0	0	0	0	0	0	0	1	1	1	0	0	0
N10	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0
N11	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0
N12	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1
N13	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
N14	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
N15	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
N16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
N17	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
N18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
N19	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
N20	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
GND	0	0	1	1	0	0	0	1	1	0	0	0	1	1	0

Fig. 15. Node-component incidence matrix of the third-order ladder network.

Based on the extracted topology and mapped parameter labels, the Kirchhoff equations for unit i are generated as:

$$V_i - V_{i+1} = R_{si}I_i + L_{ii}\dot{I}_i + \sum_{k \neq i}^n M_{ik}\dot{I}_k \quad (22)$$

$$I_i - I_{i+1} = K_i(\dot{V}_{i+1} - \dot{V}_i) + C_{i+1}\dot{V}_{i+1} + K_{i+1}(\dot{V}_{i+1} - \dot{V}_{i+2}) \quad (23)$$

where R_{si} and L_{ii} represent the resistance and leakage inductance of the stator winding, respectively, and M_{ik} denotes the mutual inductance between different units.

For this third-order ladder network, the generated parametric matrices are:

$$\mathbf{R} = \begin{bmatrix} 0 & R_{s1} & 0 & 0 & 0 \\ 0 & 0 & R_{s2} & 0 & 0 \\ 0 & 0 & 0 & R_{s3} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{L} = \begin{bmatrix} 0 & L_{11} & M_{12} & M_{13} & 0 \\ 0 & M_{12} & L_{22} & M_{23} & 0 \\ 0 & M_{13} & M_{23} & L_{33} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{T} = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} K_1+C_1 & -K_1 & 0 & 0 \\ -K_1 & K_1+K_2+C_2 & -K_2 & 0 \\ 0 & -K_2 & K_2+K_3+C_3 & -K_3 \\ 0 & 0 & -K_3 & K_3+C_4 \end{bmatrix} \quad (24)$$

By substituting the generated parametric matrices into the descriptor state-space model, the voltage-gain response of the U-phase stator winding is calculated. In this case, the voltage-gain response is selected as the target characteristic, i.e., $\mathcal{F}(s) = G(s)$, followed by constructing a fitness function (11) to minimize the discrepancy between the measured and identified voltage-gain response. The identified parameters are listed in Table V. Fig. 12 compares the identified and measured FRA signatures.

TABLE V
IDENTIFICATION RESULTS OF U-PHASE WINDING PARAMETERS

Parameters	GA	PSO	GWO
R_{s1} (m Ω)	4.1536	6.1483	0.3994
R_{s2} ($\mu\Omega$)	6.3379	3.1702	0.5089
R_{s3} (Ω)	5.3435	5.9631	3.6886
L_{11} (mH)	0.0594	0.0609	0.0459
L_{22} (mH)	1.8757	1.9636	2.0841
L_{33} (mH)	0.0300	0.0248	0.0176
M_{12} (mH)	0.0590	0.0483	0.0095
M_{13} (mH)	0.0237	0.0238	0.0176
M_{23} (mH)	0.0267	0.0132	0.0066
K_1 (nF)	6.1904	5.4490	9.8404
K_2 (nF)	5.2229	5.4060	5.1847
K_3 (nF)	12.9068	19.1689	25.5586
C_1 (nF)	6.2503	3.1536	0.0145
C_2 (nF)	3.6786×10^{-6}	2.2200×10^{-6}	2.3327×10^{-6}
C_3 (nF)	9.4712	13.2580	19.0990
C_4 (nF)	0.5103	0.7731	0.1120

TABLE VI
QUANTITATIVE EVALUATION OF IDENTIFICATION PERFORMANCE FOR THE U-PHASE STATOR WINDING

Algorithm	E_{MAPE} (%)	E_{RMSE}	E_{RP}
GA	34.77	4.8794×10^{-2}	0.6330
PSO	29.60	4.4820×10^{-2}	0.1663
GWO	29.29	4.4051×10^{-2}	0.1383

The overall identification performance is evaluated in Table VI. GWO gives the lowest values among the three algo-

rithms, with an E_{MAPE} of 29.29%, an E_{RMSE} of 4.4051×10^{-2} , and an E_{RP} of 0.1383. Due to model simplifications, such as the neglect of dielectric conductance, amplitude differences exist at the first resonance point. However, the frequencies and amplitudes of other resonance and anti-resonance points are identical, while the two FRA curves have the same trend.

C. Experimental Validation: CM and DM High-Frequency Lumped Parameter Circuit Models

This validation case is also based on the same synchronous machine stator winding described in Section III-B. The experimental setup and measurement procedure follow [19]. In high-frequency machine modeling, single-phase winding impedance measurements may be affected by coupling between adjacent phases. Therefore, common-mode (CM) and differential-mode (DM) measurement configurations are adopted to reduce interphase coupling effects and obtain broadband impedance characteristics. A Keysight E4990A impedance analyzer is used to perform sweep frequency measurements from 20 Hz to 30 MHz.

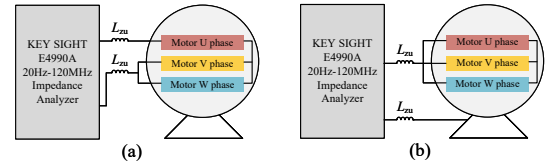


Fig. 16. CM and DM impedance measurement configurations of the synchronous machine. (a) CM impedance measurement wiring. (b) DM impedance measurement wiring.

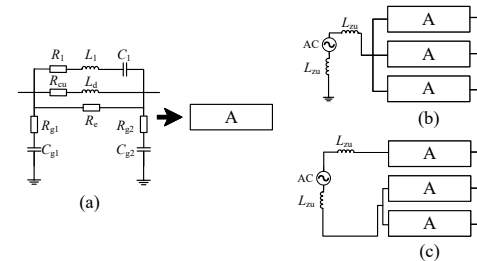


Fig. 17. High-frequency lumped parameter model based on CM and DM measurements. (a) single-phase winding model. (b) CM impedance circuit. (c) DM impedance circuit.

The CM and DM wiring schemes are shown in Fig. 16, and the corresponding high-frequency lumped parameter circuit model is shown in Fig. 17. According to Refs. [19], [41], R_{g1} and C_{g1} denote the parasitic resistance and capacitance between the stator winding and the motor frame, respectively. Similarly, R_{g2} and C_{g2} represent the parasitic resistance and capacitance between the neutral point and the motor frame. In addition, R_e , R_{cu} , and L_d represent the stator iron loss, stator copper loss, and leakage inductance, respectively. The R_1 - L_1 - C_1 branch is introduced to characterize the secondary resonance caused by inter-turn capacitance and skin effect. L_{zu} is the test lead inductance.

The CM and DM circuit diagrams in Fig. 17(b) and Fig. 17(c) are used as the inputs to the proposed framework. Fig. 18 shows the component detection and node numbering results, while the corresponding node-component incidence matrices are shown in Fig. 19(a) and Fig. 19(b), respectively. Before equation generation, the detected component labels are mapped to the parameter labels in Fig. 17. For the three identical units, $L_{i,1}$, $R_{i,1}$, $R_{i,2}$, $L_{i,2}$, and $C_{i,1}$ are mapped to L_d , R_{cu} , R_1 , L_1 , and C_1 , respectively. In addition, $R_{i,3}$, $R_{i,4}$, $R_{i,5}$, $C_{i,2}$, and $C_{i,3}$ are mapped to R_e , R_{g1} , R_{g2} , C_{g1} , and C_{g2} , respectively, and L_4 and L_5 are mapped to L_{zu} .

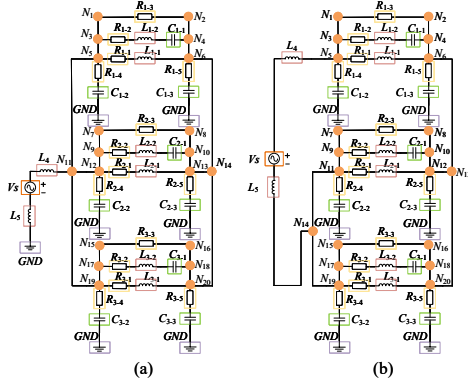


Fig. 18. Component detection and node numbering for the CM and DM circuits. (a) CM circuit. (b) DM circuit.

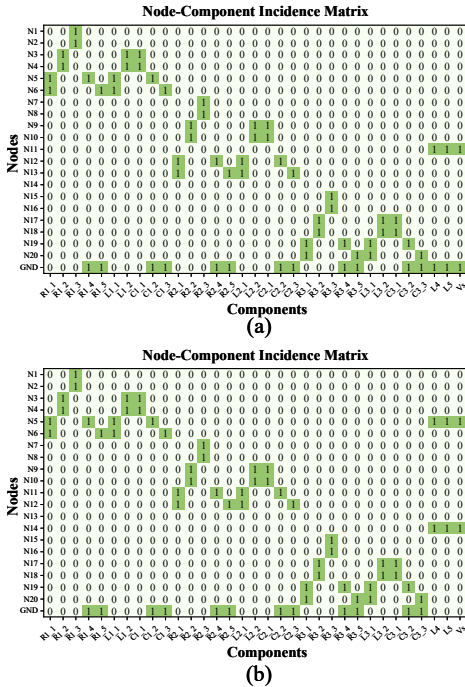


Fig. 19. Node-component incidence matrices of the CM and DM circuits. (a) CM circuit. (b) DM circuit.

Based on the node-component incidence matrices and the mapped parameter labels, the Kirchhoff equations of the CM and DM circuits are automatically generated and reorganized into descriptor state-space models. For the CM circuit, the

generated state-space model is expressed as:

$$\mathbf{A}_{cm} \begin{bmatrix} \dot{\mathbf{V}} \\ \dot{\mathbf{I}} \end{bmatrix} = \mathbf{B}_{cm} \begin{bmatrix} \mathbf{V} \\ \mathbf{I} \end{bmatrix} + \begin{bmatrix} \mathbf{P} \\ \mathbf{0} \end{bmatrix} V_s \quad (25)$$

where $\mathbf{P} = [1, 0, \dots, 0]^T$ is the input-selection vector, and $\mathbf{0}$ is a zero vector with compatible dimension. The term $\begin{bmatrix} \mathbf{P} \\ \mathbf{0} \end{bmatrix} V_s$ indicates that the source voltage V_s is injected only into the first voltage equation. The system matrices $\mathbf{A}_{cm}$ and $\mathbf{B}_{cm}$ are defined as follows:

$$\mathbf{A}_{cm} = \begin{bmatrix} R_{g1}C_{g1} & 0 & 0 & 2L_{zu} & 0 & 0 \\ -R_{g1}C_{g1} & 0 & R_{g2}C_{g2} & 0 & 0 & L_d \\ 0 & R_1C_1 & 0 & 0 & L_1 & -L_d \\ 0 & 0 & 0 & 0 & 0 & 1 \\ C_{g1} & 0 & 0 & 0 & 0 & -L_d/R_e \\ 0 & 0 & C_{g2} & 0 & 0 & -L_d/R_e \end{bmatrix}, \quad (26)$$

$$\mathbf{B}_{cm} = \begin{bmatrix} -1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & -1 & 0 & 0 & -R_{cu} \\ 0 & -1 & 0 & 0 & R_{cu} & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{3} & -1 & -(1 + \frac{R_{cu}}{R_e}) \\ 0 & 0 & 0 & 0 & 1 & 1 + \frac{R_{cu}}{R_e} \end{bmatrix}$$

For the DM circuit, the generated state-space model is given by:

$$\mathbf{A}_{dm} \begin{bmatrix} \dot{\mathbf{V}} \\ \dot{\mathbf{I}} \end{bmatrix} = \mathbf{B}_{dm} \begin{bmatrix} \mathbf{V} \\ \mathbf{I} \end{bmatrix} + \begin{bmatrix} \mathbf{P} \\ \mathbf{0} \end{bmatrix} V_s \quad (27)$$

where the system matrices $\mathbf{A}_{dm}$ and $\mathbf{B}_{dm}$ are defined as follows:

$$\mathbf{A}_{dm} = \begin{bmatrix} C_{g1}R_{g1} & 0 & 0 & 0 & -C_{g1}R_{g1} & 2L_{zu} & 0 & 0 & 0 & 0 \\ -C_{g1}R_{g1} & C_{g2}R_{g2} & 0 & 0 & 0 & 0 & L_d & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -L_d & L_1 & 0 \\ 0 & 0 & C_1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & C_1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & C_{g2}R_{g2} & 0 & 0 & -C_{g1}R_{g1} & 0 & 0 & 0 & 0 & L_d \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & L_d & -L_1 \\ C_{g1} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{L_d}{R_e} & 0 & 0 \\ 0 & 3C_{g2} & 0 & 0 & 0 & 0 & -\frac{L_d}{R_e} & 0 & -2\frac{L_d}{R_e} & 0 \\ 0 & 0 & 0 & 0 & -2C_{g1} & 0 & 0 & 0 & -2\frac{L_d}{R_e} & 0 \end{bmatrix},$$

$$\mathbf{B}_{dm} = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 & 0 & -R_{sd} & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 0 & 0 & -R_{s1} & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & -R_o \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 \\ -G_{g1} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 \\ 0 & -C_{g2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -G_{g1} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(28)

The generated descriptor state-space models are then used to calculate the CM and DM terminal response. In this case, the terminal impedance is selected as the target characteristic, i.e., $\mathcal{F}(s) = Z_s(s)$. The corresponding fitness function (11) is constructed to minimize the discrepancy between the measured and identified impedance responses. The generated models are combined with GA, PSO, and GWO for parameter identification. The identified parameters are summarized in Table VII, and the measured and identified impedance signatures are compared in Fig. 20.

The quantitative evaluation results are summarized in Table VIII. For the combined CM and DM impedance responses, GWO achieves the lowest E_{MAPE} and E_{RMSE} , with values of 18.34% and 1.5576×10^4 , respectively. GA gives the lowest E_{RP} , with a value of 3.5587 at the selected resonance and anti-resonance points. As shown in Fig. 20, the identified curves follow the measured impedance trends over a wide frequency range. Deviations mainly appear in the high-frequency region, which can be attributed to measurement noise and the limitations of the lumped parameter approximation.

TABLE VII
IDENTIFICATION RESULTS OF CM AND DM CIRCUITS

Parameters	GA	PSO	GWO
R_1 (Ω)	178.5725	120.8951	166.2580
L_1 (mH)	12.5150	10.2290	11.0700
C_1 (nF)	23.1710	22.6065	21.2955
R_{cu} (Ω)	4.0725	3.2132	2.5141
L_d (mH)	53.3743	55.3872	56.3045
R_e (Ω)	700.5600	670.5360	1020.2758
R_{g1} (Ω)	19.4797	21.4702	20.4674
C_{g1} (nF)	0.9468	0.9481	0.9515
R_{g2} (Ω)	3.5000	5.3600	5.0211
C_{g2} (pF)	8.9766	6.5883	1.8058
L_{zu} (μ H)	0.3007	0.3006	0.2998

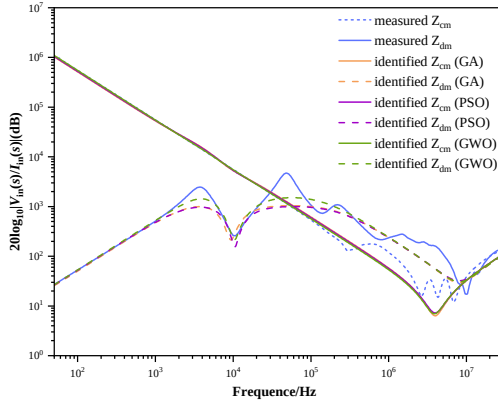


Fig. 20. Comparison between identified and measured impedance responses for CM and DM winding configurations.

IV. DISCUSSION AND LIMITATION

Currently, extensive research has been conducted on FRA-based modeling of power equipment with winding structures [2], [16], [19]–[22]. However, existing approaches still require personnel to manually derive mathematical representations from circuit diagrams before frequency-response analysis and related downstream tasks can be performed. This process is often complex, time-consuming, and prone to human error. Although AI has been increasingly applied to the power system domain [28]–[31], with DL techniques demonstrating potential in circuit diagram recognition, existing studies have mainly focused on component detection, schematic digitization, and netlist generation, without establishing a complete end-to-end framework from circuit diagrams to FRA-based mathematical representations. In addition, some studies employ DL-based surrogate models to predict transformer responses. However, these models generally provide end-to-end input-output mappings without explicitly representing the internal physical mechanisms, thereby limiting their applicability to downstream tasks requiring interpretable circuit

parameters [42], [43].

The validation results demonstrate the applicability of the proposed framework to the tested circuit topologies. Case 1 uses a fifth-order RLC lumped parameter network for simulation validation. Case 2 uses a ladder network to consider a practical synchronous machine stator winding. Case 3 further evaluates the CM and DM high-frequency equivalent circuits of the same stator winding. These cases show that the proposed framework can convert different equivalent circuit diagrams into descriptor state-space models and combine the generated models with different optimization algorithms for parameter identification. Although this study focuses on parameter identification for rapid equivalent-circuit modeling, the generated state-space models can also provide a basis for other circuit-analysis tasks, such as transient analysis and frequency response analysis.

Despite these advantages, several limitations remain. First, the present framework assumes frequency-independent circuit parameters. In practical high-frequency winding models, resistance and inductance may vary with frequency because of skin effect, proximity effect, and magnetic losses. Future work will extend the framework to frequency-dependent RLC models. Second, the current implementation mainly supports passive RLC networks. Components with more complex characteristics, such as controlled sources, nonlinear elements, and frequency-dependent branches, have not yet been incorporated. Expanding the component library and the corresponding equation-generation rules will be investigated in future work. Third, the component detection model is trained primarily on standard circuit symbols. Its robustness to hand-drawn diagrams, low-quality scanned images, and non-standard symbol representations remains limited. Future work will improve the adaptability of the detection model to different drawing styles and symbol variations.

V. CONCLUSION

This study proposes an intelligent circuit identification framework for automated modeling of power equipment with winding structures. By integrating DL-based component detection, graph-theory-based topology extraction, and circuit-theory-based equation generation, the proposed framework converts equivalent circuit diagrams into descriptor state-space models for FRA calculation and parameter identification. The generated models are further combined with different optimization algorithms to estimate circuit parameters from external terminal measurements. The effectiveness of the framework is validated through three case studies. The results demonstrate that the proposed framework can process different equivalent circuit topologies and reduce the manual effort required for equation derivation and model construction.

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